

Investigating Perception

Perceptual Completion under Occlusion:

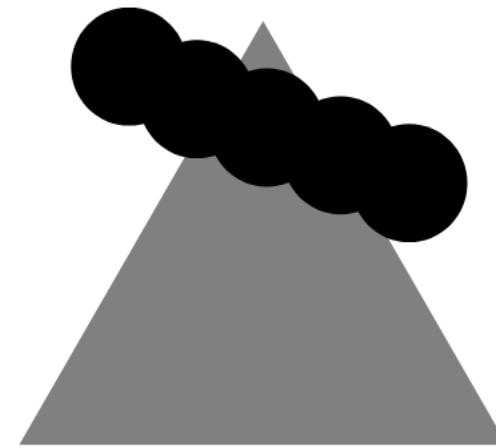
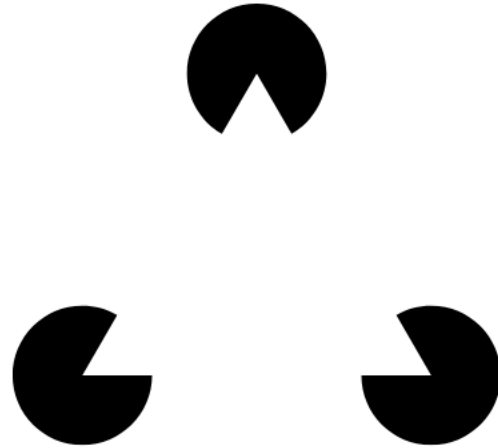
How Visual Evidence is Integrated

Bachelor Thesis Symposium – Exposé Presentation

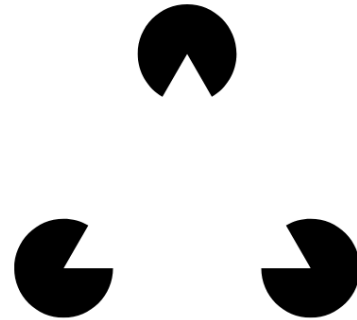
Solana Neumeier

10.06.2026

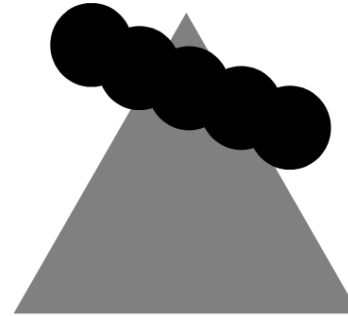
Theoretical Background: Perceptual completion



Theoretical Background: Perceptual completion – Modal and Amodal



(a) Modal completion



(b) Amodal completion

- Classic dichotomy :
- Directly visible → Modal
- Occluded → Amodal

Theoretical Background

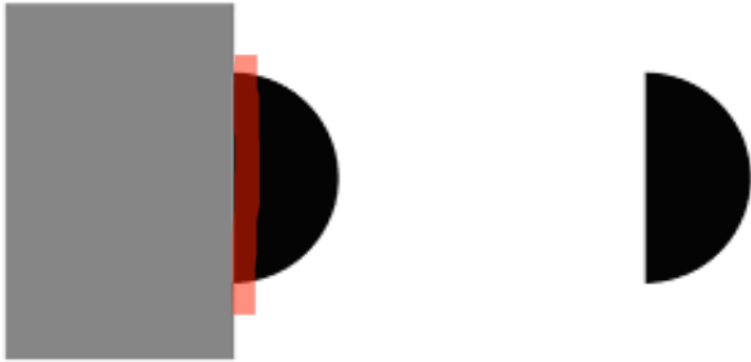
Relevant Phenomenon: Occlusion Illusion



The directly visible part of a partially occluded object appears larger than a physically identical region that is fully visible and not occluded.

Theoretical Background

Relevant Phenomenon: Occlusion Illusion and PMC



The directly visible part of a partially occluded object appears larger than a physically identical region that is fully visible and not occluded.

Partial modal completion hypothesis: occlusion cues lead to amodal completion, while parts of the completed region are still experienced as modal

Theoretical Background

Relevant Concepts: new criteria for Modal Amodal



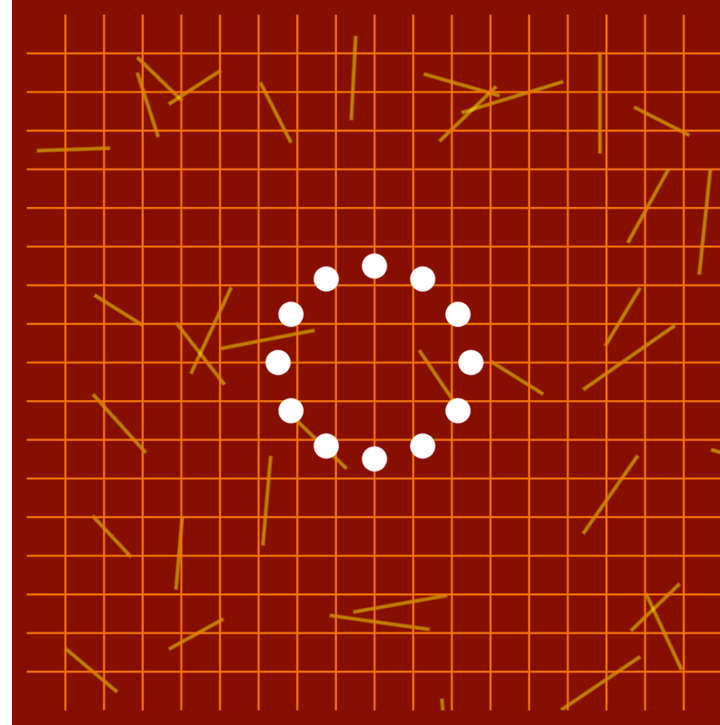
Palmer : conclusiveness of sensory evidence as a continuum



Previous Work

Bachelor Thesis Christian Haase 2025

- Scherzer & Ekroll (2015) Real-world replication
- Stimuli :



Previous Work

Bachelor Thesis Christian Haase 2025

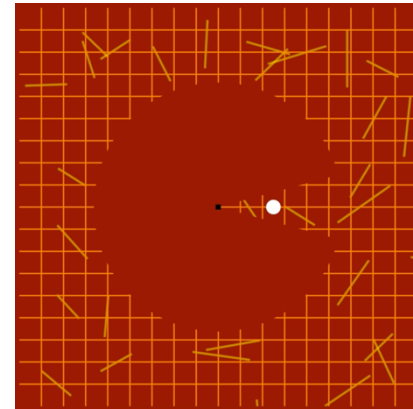
Experiment – Methods:

Independent variable

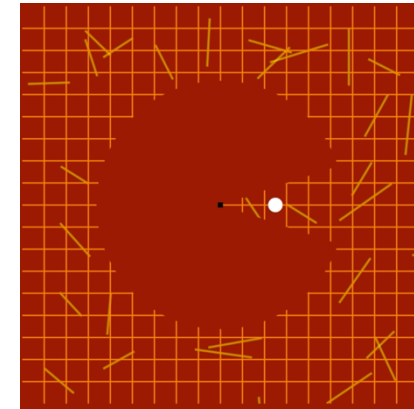
- Opening angle of occluder (30, 36.9, 60, 66.9, 90 deg)

Dependent variable

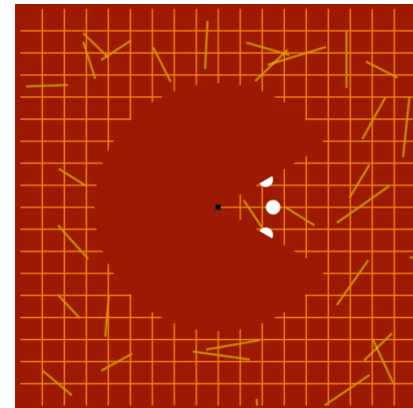
- Perceived # dots (counting task)



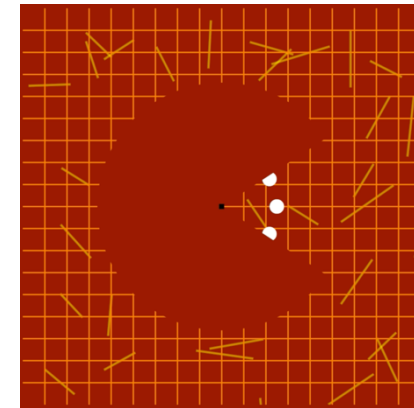
30°



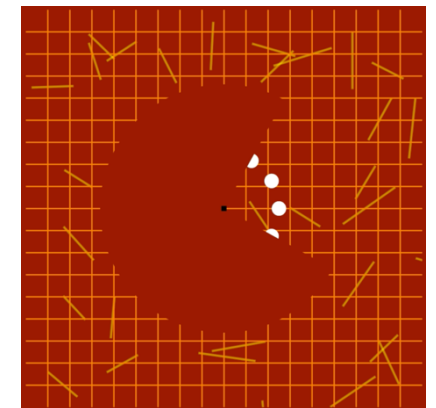
36.9°



60°



66.9°



90°

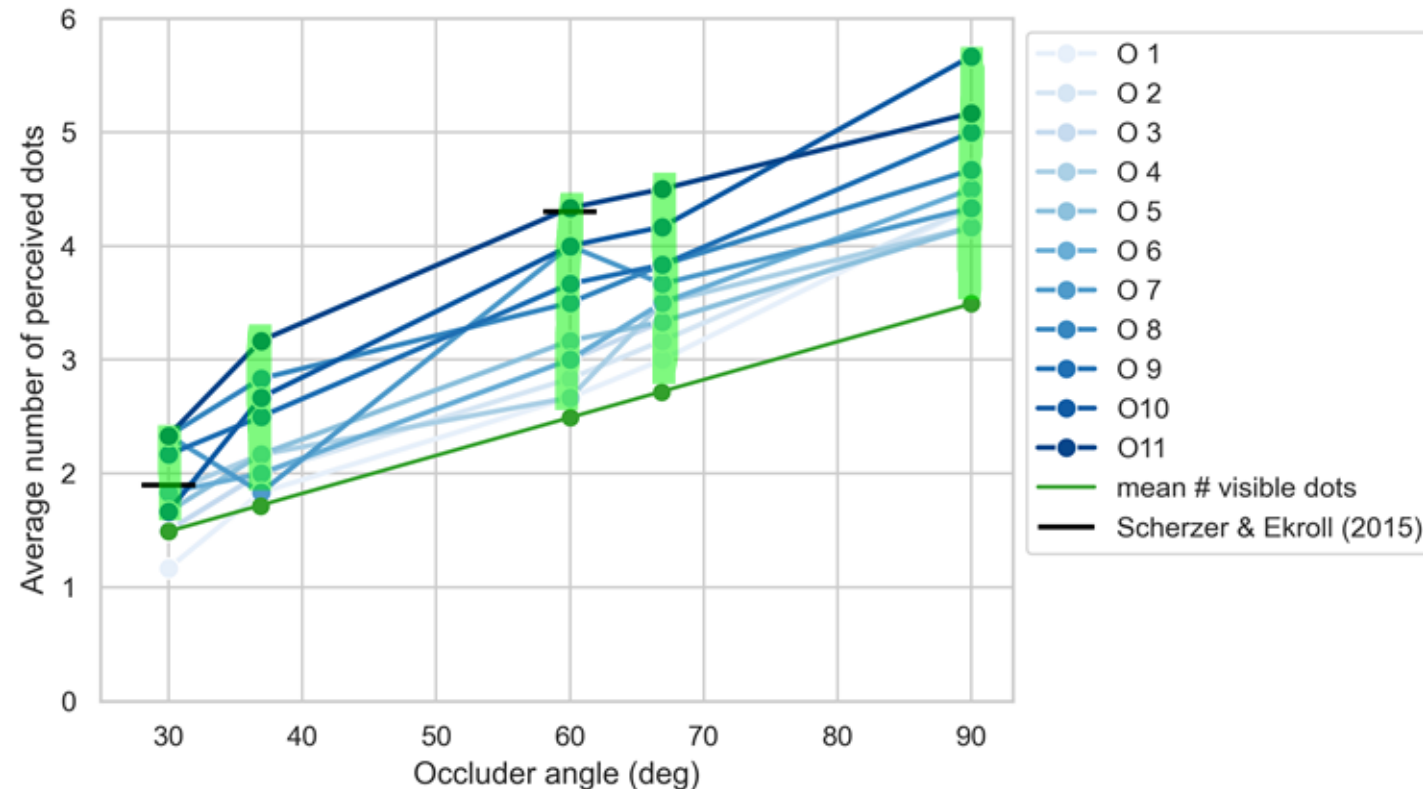
Previous Work

Bachelor Thesis Christian Haase 2025

Experiment - results:

→ # dots perceived higher
than # dots physically visible
76,25

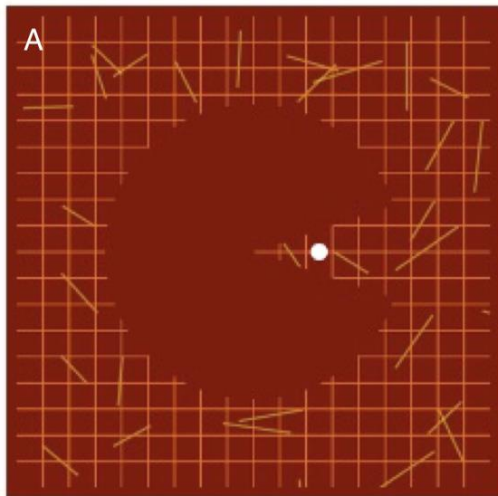
→ real-world stimulus produces
similar effect to digital experiment



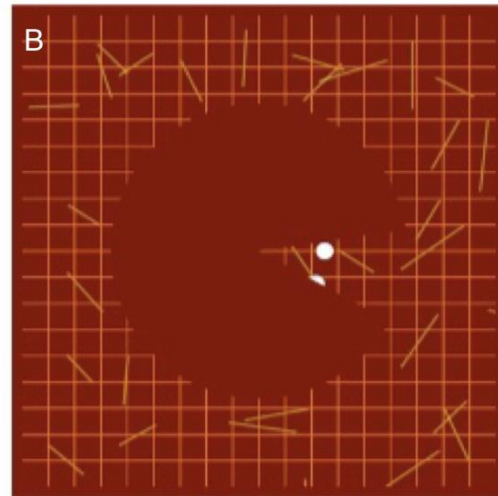
Motivation

how visual information relates to perceived number of dots

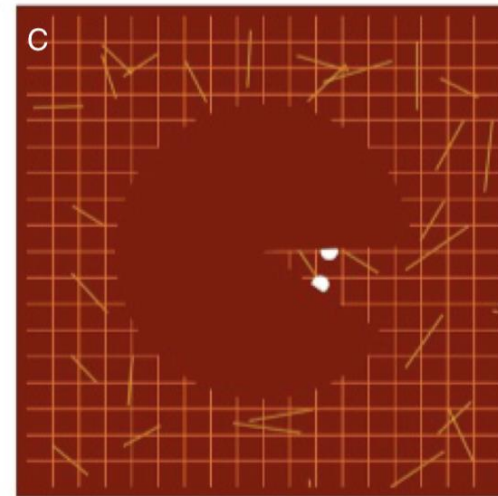
Example :
36.9° occluder



Single dot
in 110 frames



single dot
+ partially visible dot
in 150 frames

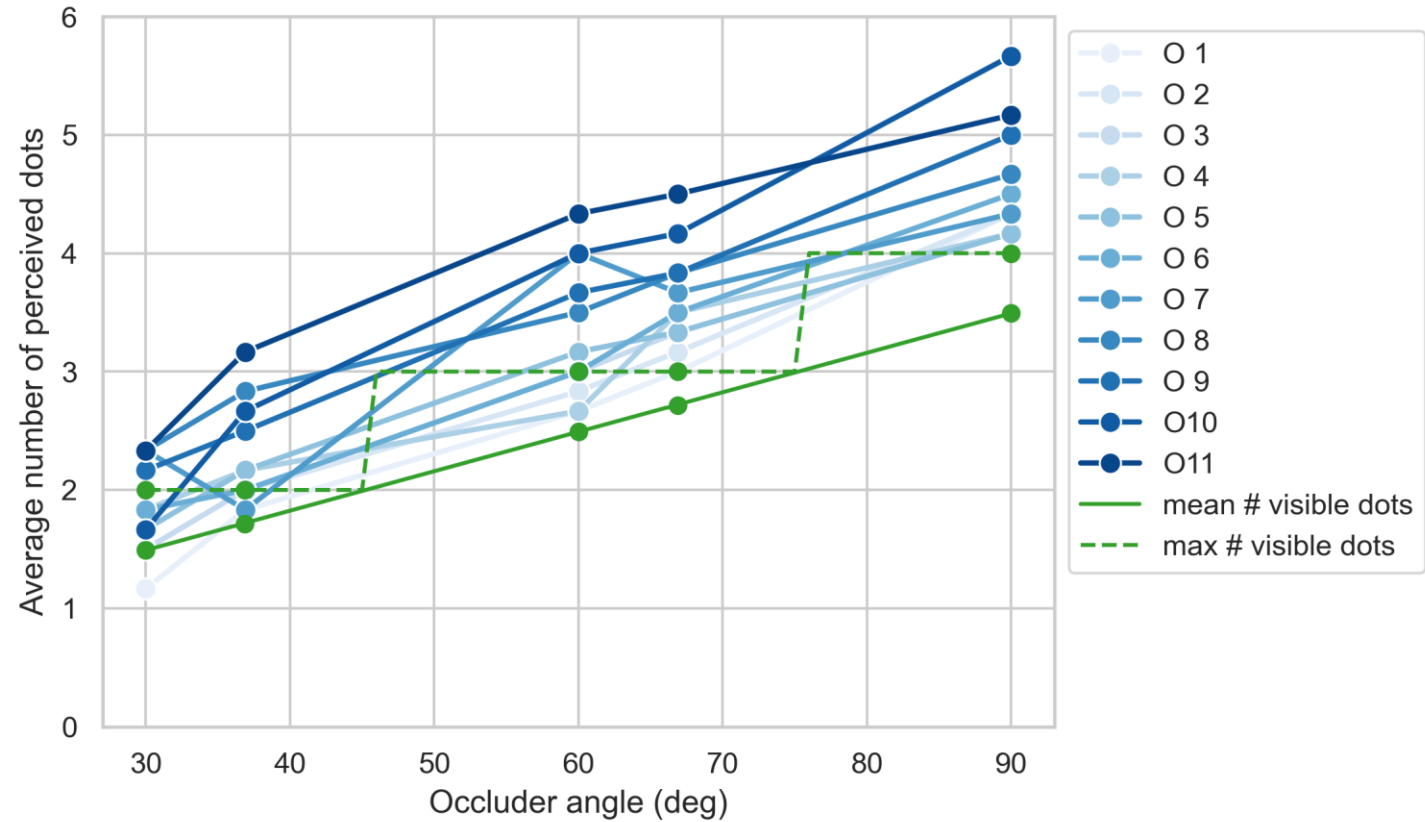


two partially visible dots
in 100 frames

! Partially visible dot = 1 dot

-> mean # visible dots = 1.694
max # visible dots = 2

Motivation



dot spacing = $360/12 = 30^\circ$

dot size $\approx 15^\circ$

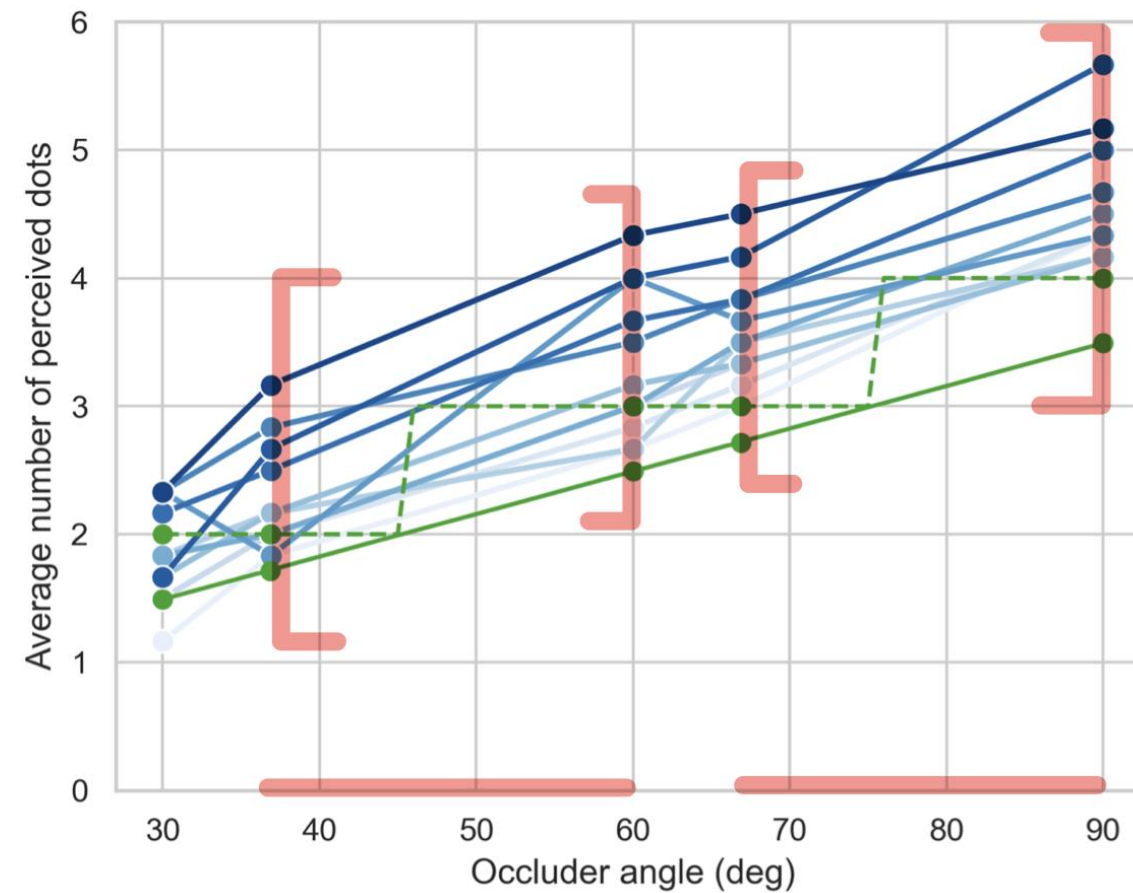
Research question

How do observers report the perceived number of dots in the perceptual completion paradigm? Does it relate to the maximum number of visible dots, the mean number of visible dots, or follow a different pattern?

Hypothesis

If perception follows the maximum number of visible dots, the perceived number of dots is expected to show step-like increases at opening angles around 45° and 75° .

Aim



Methode : Experiment

Experiment plan:

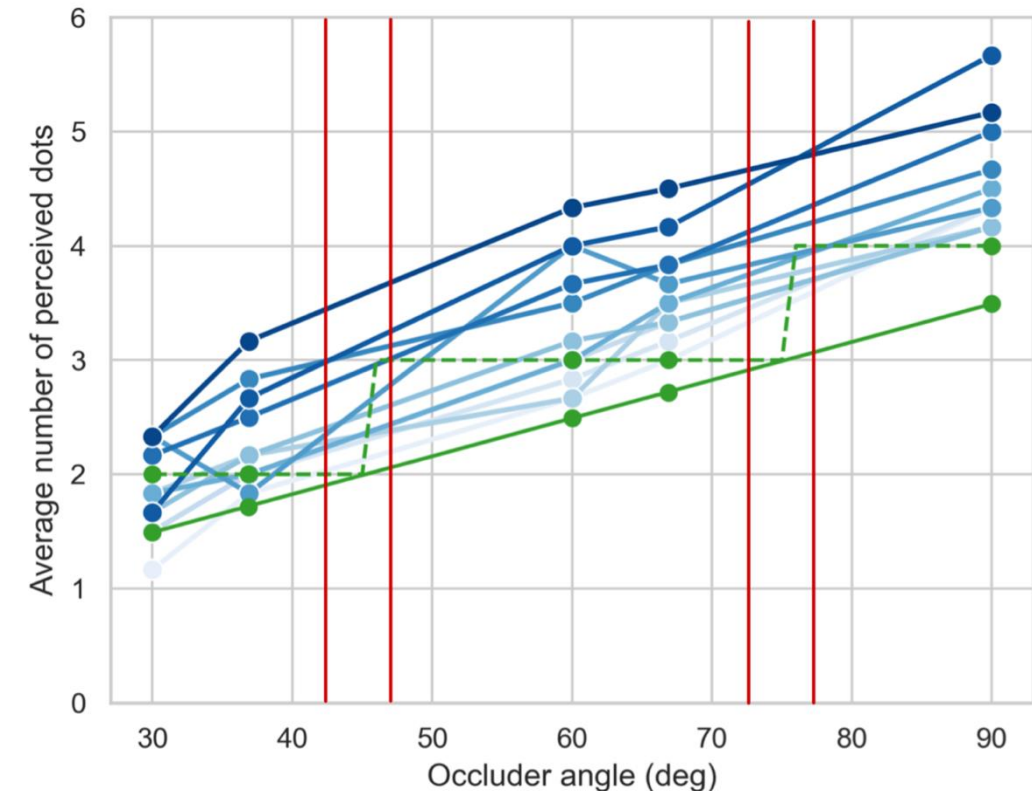
Replication of real world Haase experiment

Independent variable

- Opening angle of occluder
(30, 43, 49, 60, 73, 79, 90deg)

Dependent variable

- Perceived # dots (counting task)



Methode : Experiment

Experiment plan:

- ≥ 10 participants
- Stimuli distance 80 cm
- Answer : entering number on keyboard ; confirm by pressing enter 2x
- (unsure $\rightarrow$ range ± 1)
- Partly visible dot = 1 dot
- No time limit

Next steps:

- work out the method in more detail
- exposé
- Register thesis
- Plan the experiment
- Conduct the experiment
- Evaluate